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RESEARCH ARTICLE

Mitigating Device Heterogeneity for Enhanced Indoor Positioning System Performance Using Deep Feature Learning

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ABSTRACT Indoor Positioning Systems (IPS) are essential for delivering accurate location-based services in environments where Global Navigation Satellite Systems (GNSS) are ineffective. The Received Signal Strength Indicator (RSSI)–based IPS leverages the existing access point infrastructure to provide a cost-effective solution for indoor location determination. However, device heterogeneity, characterized by variations in hardware, sensors, and software between devices, poses a significant challenge, often degrading positioning accuracy and robustness. This study investigates the impact of device heterogeneity on IPS performance using the TUJI1 dataset, which comprises RSSI measurements collected from five different devices. After comprehensive preprocessing of the RSSI signals, the study proposes a hybrid Convolutional Neural Network–Long Short-Term Memory (CNN–LSTM) framework, leveraging advanced feature extraction techniques to improve positioning accuracy and mitigate the effects of device variability. In this framework, RSSI fingerprints are represented as two-dimensional (2D) images, enabling the CNN component to capture spatial co-occurrence patterns among access points, while the LSTM component models inter–access point dependencies, resulting in robust, device-agnostic feature embeddings. The proposed model achieves a three-dimensional (3D) mean positioning error of 2.20 m, outperforming traditional k-Nearest Neighbors (k-NN) methods. In cross-device evaluations, the model demonstrates improved robustness, reducing positioning errors by up to 0.17 m compared to conventional approaches. These results highlight the effectiveness of the CNN–LSTM architecture in addressing device heterogeneity, offering a scalable and efficient solution for diverse IPS environments. This study advances the field of indoor positioning by providing a robust framework capable of maintaining high accuracy in the presence of device variability, thus contributing to the development of more reliable and adaptable IPS technologies.

INDEX TERMS Indoor positioning, indoor localization, device heterogeneity, CNN-LSTM, feature extraction, dimensionality reduction, deep feature learning.

I. INTRODUCTION

Indoor Positioning Systems (IPS) play a critical role in delivering accurate and reliable location services in environments where Global Navigation Satellite Systems (GNSS) are ineffective [1]. Device heterogeneity—arising from variations in hardware, sensor precision, and software configurations—remains a major challenge that degrades IPS performance [2], [3]. IPS applications span diverse domains, including indoor navigation, asset tracking, and intelligent building management, where precision and reliability are essential. Despite significant advancements, device-induced

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inconsistencies continue to constrain IPS performance [2], [4].

Addressing device heterogeneity is essential to unlocking the full potential of indoor localization systems. Variations in device hardware, antenna design, and signal processing capabilities introduce inconsistencies in signal measurements, leading to localization errors and reduced system robustness [5]. Researchers have addressed these challenges by employing advanced techniques such as Deep Learning (DL) and Transfer Learning (TL), which enhance both accuracy and scalability in indoor localization (IL) systems [6], [7], [8], [9]. Comprehensive analyses of machine learning (ML) techniques highlight the rapid evolution of algorithms in wireless localization, reinforcing the pivotal role of DL in improving IPS robustness [10]. Traditional approaches often struggle to handle signal fluctuations and environmental variability, motivating the development of more robust datasets and evaluation metrics to ensure reliability as IL applications expand into areas such as asset tracking, energy management, and emergency evacuation. AI-driven methodologies, including ML and hybrid models, have thus emerged as promising solutions to these challenges. With the advent of 6G networks, researchers further anticipate substantial gains in IL performance and scalability, underscoring the need for adaptable and efficient positioning frameworks [11].

One of the most widely studied approaches to IL involves Wi-Fi signal-based positioning, where DL models have demonstrated superior performance over traditional machine learning methods in handling signal fluctuations and improving accuracy [12]. This process typically comprises three stages: data preprocessing, feature extraction, and model development. Despite the widespread reliance on the Received Signal Strength Indicator (RSSI) as the primary metric for Wi-Fi-based positioning, challenges such as signal volatility and dataset limitations persist. Long Short-Term Memory (LSTM) networks mitigate these issues by stabilizing signal variations and enhancing robustness, rendering them highly effective for complex positioning tasks [13]. Device heterogeneity further complicates RSSI-based localization, as hardware variations introduce inconsistencies in signal strength. To address these challenges, researchers have proposed innovative solutions, including Gaussian process regression for joint calibration and adversarial training through neural networks, both of which achieve higher localization accuracy than conventional techniques [14].

Researchers have also explored Wi-Fi fingerprint crowdsourcing and the integration of unlabeled fingerprints as practical strategies to enhance localization performance. These approaches underscore the importance of maintaining high accuracy across diverse hardware configurations. Hybrid architectures, such as Convolutional Neural Network–Long Short-Term Memory (CNN–LSTM) models, have demonstrated strong potential in mitigating device-related inconsistencies, highlighting their relevance in modern indoor localization systems [14]. Complementary studies combining deep learning with Wi-Fi fingerprinting have further shown improved matching performance, reinforcing the utility of hybrid models for robust localization [13], [15]. Moreover, recent semi-supervised augmentation techniques, including MoDeFA, demonstrate the capability to reconstruct robust fingerprints from limited labeled data, thereby reducing the reliance on costly site surveys [16]. Collectively, ML-based solutions have proven effective in addressing non-line-of-sight (NLOS) issues and accommodating device heterogeneity in indoor positioning.

Both supervised and unsupervised machine learning techniques have significantly advanced fingerprinting-based localization and data fusion strategies, enhancing positioning accuracy and emphasizing the importance of sophisticated architectures such as CNN–LSTM for addressing device heterogeneity [17]. Public Wi-Fi fingerprint datasets have further facilitated IPS research; however, challenges related to data dispersion, inconsistent organization, and limited accessibility remain [18]. Dense, small-scale testbeds often achieve superior performance compared to larger datasets with sparse sampling, underscoring the need for standardized formats and structured data collection. For example, hierarchical Bayesian models that account for transmitter and receiver variability have reduced localization errors to a median of 1.7 m with a 95% confidence interval, demonstrating the critical role of hardware variability in accurate positioning [19]. These findings reinforce the necessity of considering device heterogeneity in model design, further supporting the adoption of CNN–LSTM architectures to mitigate device-specific inconsistencies in IL systems.

Scholars have extensively reviewed the application of machine learning techniques in indoor positioning, emphasizing accuracy, scalability, and computational efficiency. Multipath fading and NLOS conditions continue to present significant challenges to achieving high-precision localization. Comparative analyses of input measurement techniques, such as RSSI-based fingerprinting and time-based methods, underscore the importance of selecting appropriate data sources to optimize performance. These insights further support the adoption of advanced hybrid models, such as CNN–LSTM, to effectively address these challenges and enhance localization robustness [20].

Beyond accuracy, privacy preservation has emerged as a critical consideration in indoor localization systems. Researchers have explored secure computation methods, including two-party computation and homomorphic encryption, to safeguard sensitive localization data [21]. Given the widespread adoption of Wi-Fi technologies for indoor positioning, ensuring privacy without compromising performance remains an essential design objective. This growing emphasis on adaptability and privacy further reinforces the relevance of CNN–LSTM models, which contribute to improved localization reliability in privacy-sensitive environments [22].

Recent advances in hybrid deep learning architectures have further strengthened IL capabilities. CNN–LSTM models have demonstrated superior performance over baseline techniques by enhancing feature extraction and prediction

accuracy, particularly under conditions of signal distortion and device diversity [23], [24]. These findings underscore the effectiveness of CNN–LSTM architectures in addressing device heterogeneity and improving robustness in real-world IL applications.

To enhance interpretability, recent frameworks have integrated explainable AI (XAI) techniques such as Local Interpretable Model-agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP), providing valuable insights into model decision processes [25]. Beyond Wi-Fi-based localization, multisensory positioning solutions now leverage data fusion from diverse technologies and sensors. For instance, Asaad et al. [26] proposed Wi-Lo, a system that integrates Wi-Fi and LoRa RSSI to enable seamless indoor–outdoor transitions, while Fryad et al. [27] combined smartphone magnetometer readings with Wi-Fi RSS to mitigate multipath effects and device-specific biases. Collectively, these advancements highlight the growing importance of hybrid architectures and sensor fusion for achieving robust, interpretable, and scalable localization systems.

Researchers recently introduced the TUJI1 dataset [28] to study measurement granularity and device heterogeneity in IL systems. Analysis of TUJI1 shows that device performance variations significantly affect localization accuracy, underscoring the need for systematic data collection procedures. Designed to support diverse ML models, TUJI1 serves as a valuable resource for developing robust localization techniques. These findings highlight the critical role of high-quality datasets in addressing device heterogeneity and reinforce the importance of structured data collection for CNN–LSTM-based positioning models [28]. Beyond CNN–LSTM models, researchers have explored additional approaches to enhance IL accuracy. For example, introducing the Intersection over Union (IoU) metric [30] improved the generalization of k-NN-based systems, enabling better performance across multiple datasets. When supplemented with appropriate metrics, k-NN models retain simplicity and robustness, making them viable in specific scenarios [30]. Similarly, a signal statistics-based universal fingerprinting solution leverages statistical analysis to optimize hyperparameters, reduce computational overhead, and accommodate diverse device profiles while maintaining consistent performance across datasets [31]. Researchers have also integrated inertial sensors with ML classifiers to address device heterogeneity in IL systems. A wearable sensor-based approach that applies preprocessing techniques such as Chebyshev filtering and feature extraction has achieved high localization accuracy, demonstrating the effectiveness of advanced preprocessing pipelines in mitigating device inconsistencies [32]. Furthermore, studies on Informative Path Planning (IPP) frameworks optimize survey paths to improve data collection efficiency while preserving localization accuracy. By reducing redundant information, these frameworks help mitigate data disparities caused by heterogeneous devices [33].

A recent study introduces the rapid fingerprint collection mechanism TuRF for RSSI-based localization, employing Gaussian Process Regression (GPR) to compensate for signal variations across different hardware. By emphasizing fast data collection, TuRF provides a scalable solution for heterogeneous environments [34]. Yildirim et al. [35] also propose a dimension reduction technique for RSSI-based indoor positioning that reduces computational complexity through Manhattan distance calculations and feature selection while enhancing positioning accuracy. Their method further standardizes input data from diverse devices, ensuring consistent localization results.

These studies highlight the need to integrate advanced algorithms, rigorous preprocessing, and adaptive metrics to overcome hardware heterogeneity. Building on this foundation, our study proposes a unified CNN–LSTM framework that (i) corrects device-level inconsistencies during fingerprint preprocessing, (ii) jointly learns spatial and inter-AP correlations to improve cross-device generalization, and (iii) maintains real-time feasibility in dynamic indoor environments.

This study investigates the impact of device heterogeneity on the performance of IPS and proposes a hybrid CNN–LSTM framework, with the following research questions and contributions:

Research questions:

- How does device heterogeneity influence the accuracy and robustness of RSSI-based Indoor Positioning Systems?
- To what extent can a hybrid CNN–LSTM model mitigate the adverse effects of device heterogeneity and enhance positioning accuracy compared with traditional k-Nearest Neighbors methods?
- How effectively does the CNN–LSTM framework generalize across heterogeneous devices in cross-device scenarios, ensuring robust performance on unseen hardware configurations?

Contributions:

- Developed a CNN–LSTM model that achieved a 3D positioning error of 2.20 m, surpassing conventional k-NN methods.
- Enhanced cross-device generalization, reducing positioning errors by up to 0.17 m compared with k-NN.
- Designed a comprehensive preprocessing pipeline to remove duplicated coordinates and identify undetected access points, enhancing data integrity.
- Advanced IPS benchmarking standards by utilizing the TUJI1 dataset for reproducible and transparent performance evaluation.

The proposed approach enhances IPS robustness by eliminating redundant data and employing advanced deep feature extraction through CNN and LSTM networks to analyze the influence of device characteristics on positioning accuracy. Utilizing the high-density TUJI1 dataset with data points spaced 40–45 cm apart, this work reveals that

more powerful devices do not necessarily yield superior positioning outcomes, informing the development of scalable and equitable IPS solutions.

Figure 1 illustrates the comprehensive workflow adopted for processing the TUJI1 dataset using the proposed hybrid CNN–LSTM framework. The process begins with the raw dataset, followed by Exploratory Data Analysis (EDA) to examine data characteristics. Subsequent preprocessing steps include handling missing values, rectification, normalization, and the removal of duplicate entries. In parallel, the workflow performs detailed analyses such as invalid RSSI computation, fingerprint sample distribution assessment, identification of undetected access points, and determination of coordinates corresponding to the maximum and minimum numbers of detected access points. These metrics are further evaluated across devices alongside their spatial coordinates. This structured methodology ensures thorough data investigation and preparation, thereby enhancing the reliability and accuracy of the proposed framework.

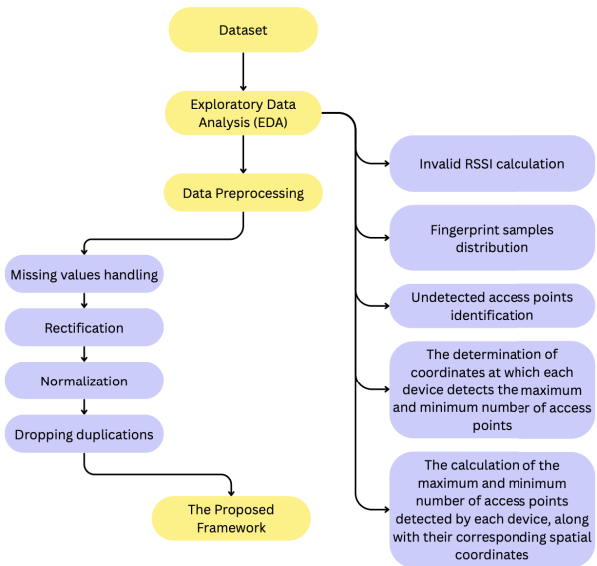

FIGURE 1. The workflow diagram.

The remainder of this paper is structured as follows: Section II introduces the dataset, detailing its sources and the distribution of sample instances. Section III describes the data preprocessing procedures, including the handling of missing values, normalization, rectification, and duplicate removal, and further elaborates on the proposed CNN–LSTM model architecture, outlining its layers, hyperparameters, and implementation details. Section IV presents the experimental setup, evaluation metrics, and performance results of the proposed approach. Finally, Section V concludes the paper and discusses future research directions.

II. DATASET

The research employs the TUJI1 dataset [28] for its high data density and comprehensive coverage of device diversity. This comprehensive dataset is well-suited for examining how device diversity affects the efficacy of indoor positioning systems. TUJI1 encompasses information gathered from five distinct devices, with data points spaced no more than 45 cm apart, ensuring a high level of granularity. However, the fine granularity may be affected by potential inaccuracies stemming from manual surveying, which could introduce errors due to imprecise device positioning, human body blockage, and surveyor height variations, unlike automated methods in datasets like the Microsoft Indoor Localization Competition (MILC).

Devices capture Wi-Fi RSSI readings from 310 access points distributed at unknown locations within the deployment area. The presence of such a high number of access points likely causes signal interference, potentially compromising the reliability of RSSI measurements. These measurements are taken across a detailed grid. Each fingerprint, consisting of RSSI values from all access points recorded by a specific device at a particular location, is associated with the exact spatial coordinates. These coordinates are determined using a BOSCH GLM 120 C laser distance meter. Despite the precision of the laser measurements, the location recording process involves a 10-second measurement duration per location, with only the last 3 seconds considered for matching RSSI values to coordinates, which may risk mismatches due to rapid signal fluctuations. The 10-second measurement duration may also not capture sufficient temporal variability, contrasting with longer sampling in UJIIndoorLoc [29]. Notably, access points 282, 283, and 284 are consistently undetected across all devices in the TUJI1 dataset, with all corresponding RSSI values recorded as 100 dBm, indicating invalid measurements.

The collection consists of 8,899 fingerprint samples, each associated with exact location coordinates. Following Klus et al. [28], the first spatial grid is fully assigned to training, while the second is randomly split 50:50 between training and testing. This ensured broad spatial coverage in the training data and randomness in the testing distribution, resulting in a final train/test split of 76:24 due to the slightly lower sample count in the second grid. Importantly, this study preserves the original dataset partitioning strategy to maintain consistency and enable direct comparability with prior methods. Additional details regarding the number of valid samples collected by each device for training and testing are presented in Table 1. Figure 2 visually represents the proportional distribution of fingerprint samples across different devices.

TABLE 1. Distribution of fingerprint samples across devices.

Devices	Samples		
	Training	Testing	Total
Galaxy S20	1408	420	1828
Galaxy S7	1336	420	1756
POCO X3	1318	429	1747
Galaxy Tab S7	1377	458	1835
Galaxy A12	1313	420	1733

The number of detected access points at a given location varies due to multiple environmental and device-specific factors. Signal propagation is influenced by multipath fading, interference, and obstructions, which cause fluctuations in RSSI and the number of visible APs at different positions, even for the same device [36]. Additionally, differences in hardware components such as antennas and signal processing techniques among various devices contribute to these variations [37]. Table 2 depicts the minimum and maximum detected access points for each evaluated device and their associated spatial coordinates.

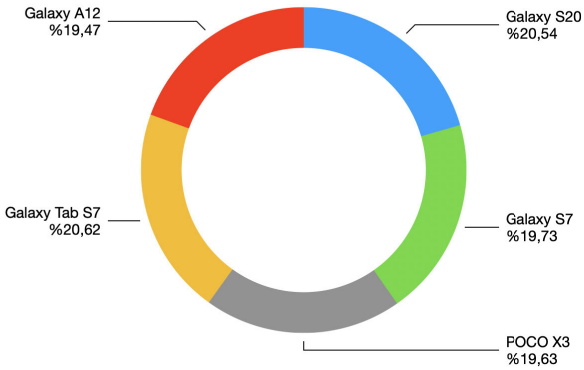

FIGURE 2. Total samples distribution across devices based on proportional representation.

TABLE 2. Access point (AP) detection variability across devices.

Device	APs		Coordinates (z=0)			
	Min	Max	X_{min}	Y_{min}	X_{max}	Y_{max}
Galaxy S20	14	76	5.958	1.193	0.2985	18.783
Galaxy S7	10	44	10.724	8.944	0.893	19.974
POCO X3	8	44	11.321	1.789	0.0	26.827
Galaxy Tab S7	2	25	2.384	5.366	9.238	16.3955
Galaxy A12	10	45	0.893	4.472	7.45	32.4935

Figures 3 and 4 depict the distribution of devices across the deployment area, highlighting the locations where the minimum and maximum numbers of access points are detected, respectively. Despite the variation in the devices' ability to detect weaker RSSI signals from more distant access points, there is also observable variation in the locations where each device detected the highest and lowest number of access points. Furthermore, since the positions of the access points are unknown, it is impossible to conclusively determine which device provided the most accurate logical location.

The deployment area where the data is collected spans a relatively large area of 32.8 m by 22.6 m. Consequently, receiving signals from all access points at a specific location is often impractical, resulting in fingerprints containing many invalid RSSI samples. Notably, the number of (#) valid RSSI measurements per fingerprint for each device reflects the device's ability to capture weaker signals. A higher number of valid RSSI values indicates a more capable and powerful device. Table 3 presents # valid and invalid RSSI

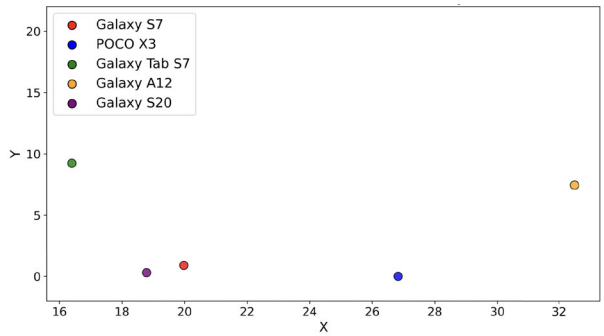

FIGURE 3. Maximum detected access points location for each device.

samples collected by each device and the percentage of invalid samples relative to the total RSSI samples. Among the devices evaluated, Galaxy S20 stands out as the most efficient, exhibiting the lowest rate of invalid RSSI samples compared to the other devices.

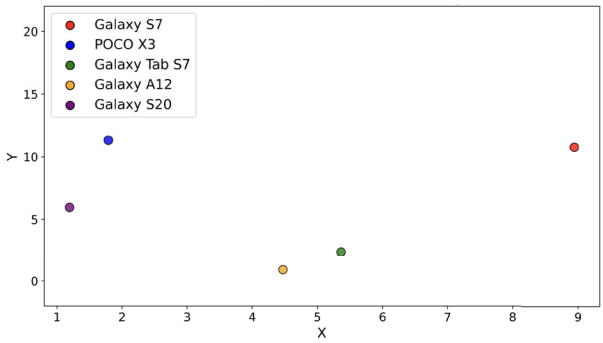

FIGURE 4. Minimum detected access points location for each device.

TABLE 3. Summary of valid and invalid RSSI samples per device.

Devices	All RSSI (#)	Invalid RSSI	
		(#)	(%)
Galaxy S20	566,680	**505,209**	**89.15**
Galaxy S7	544,360	504,182	92.62
POCO X3	541,570	504,976	93.24
Galaxy Tab S7	568,850	550,710	96.81
Galaxy A12	537,230	496,988	92.51

III. PROPOSED METHOD

The proposed deep feature learning pipeline is designed to address device heterogeneity through a series of integrated stages. Preprocessing operations—including invalid value imputation, rectification, normalization, and duplicate removal—standardize the input data and enhance resilience to device-specific irregularities. RSSI fingerprints are then transformed into structured two-dimensional (2D) matrices, enabling convolutional operations that capture spatially localized co-occurrence patterns among access points. A convolutional module extracts these spatial features, reducing dimensionality and mitigating noise, while the subsequent LSTM module captures structured inter–access point (inter-AP) dependencies, serving as a data-driven smoothing

mechanism that mitigates transient fluctuations induced by environmental or hardware variability. This hybrid representation yields discriminative, device-agnostic embeddings mapped to spatial coordinates through lightweight regression heads. The dual-stage design enables the framework to learn both local and global spatial patterns, effectively mitigating the impact of device variability.

In terms of scalability, the proposed pipeline offers two principal advantages. First, by replacing raw fingerprint matching with compact feature embeddings, it substantially reduces computational complexity and storage overhead compared to conventional k-NN approaches, thereby enabling efficient real-time inference even in large-scale deployments. Second, the incorporation of multi-device training improves generalization, as demonstrated in cross-device evaluations where the framework maintained high accuracy across unseen hardware. Collectively, these design attributes establish a scalable methodology capable of accommodating extensive datasets, heterogeneous device profiles, and potential extensions to multimodal sensor fusion.

A. DATA PREPROCESSING

The dataset undergoes a rigorous preprocessing pipeline to enhance data quality and ensure suitability for model training. This process includes imputing missing values, applying rectification, normalizing signal strength distributions, and removing duplicate coordinates.

1) MISSING VALUES HANDLING

The dataset contains valid signal strength measurements ranging from -35 to -100 dBm. Invalid RSSI values are assigned a value of 100 dBm to clearly distinguish them from weak but valid signal readings. Missing values are imputed by replacing them with the minimum RSSI value minus one, as defined by Equation 1.

$$RSSI_{invalid} = RSSI_{min} - 1 \tag{1}$$

$RSSI_{invalid}$ represents the invalid $RSSI$ values, while $RSSI_{min}$ denotes the minimum RSSI value in the dataset.

2) RECTIFICATION

The RSSI measurements in the dataset are inherently negative, which may cause complications when using the Rectified Linear Unit (ReLU) activation function. Since ReLU outputs zero for all negative inputs, potentially informative signal variations could be suppressed, leading to the dying ReLU phenomenon and consequently degrading model performance [38]. To address this issue, a rectification transformation is applied to shift all RSSI values to a positive scale, as defined in Equation 2.

$$RSSI_{rectified} = RSSI - RSSI_{min} \tag{2}$$

where $RSSI_{rectified}$ represents the transformed $RSSI$ values and $RSSI_{min}$ denotes the minimum RSSI value in the dataset.

3) DATA NORMALIZATION

The dataset is normalized to the range [0, 1] to mitigate the vanishing gradient problem and promote stable model training [39]. This normalization step is particularly important because, without scaling, gradient descent may converge unevenly when input features have differing magnitudes, resulting in imbalanced weight updates and inefficient learning. By rescaling all features to a uniform range, the optimization process becomes more stable and effective. The normalization is performed according to Equation 3.

$$RSSI_{norm} = \frac{RSSI - RSSI_{min}}{RSSI_{max} - RSSI_{min}} \tag{3}$$

where $RSSI_{norm}$ represents the normalized values, and $RSSI_{min}$ and $RSSI_{max}$ are the minimum and maximum values of the variable $RSSI$, respectively.

4) DROPPING DUPLICATIONS

In RSSI-based IPS, each fingerprint is expected to correspond uniquely to a specific location, establishing a direct relationship between signal characteristics and spatial positions. Duplicate entries can distort model training, particularly when multiple fingerprints share identical coordinates. Moreover, redundant records in training and testing datasets may lead to erroneous predictions and compromised generalization. Consequently, removing duplicate entries is a crucial step to ensure the robustness of the model and the reliability of the results.

During the EDA of the TUJI1 dataset, specifically focusing on data from the Galaxy S20 and Galaxy Tab S7 devices, multiple fingerprint samples were observed at identical coordinates. Although these duplicates share the same coordinates, they exhibit variations in RSSI measurements and the number of detected access points. For instance, as shown in Table 4, several entries for the Galaxy Tab S7 at coordinates (x = 11.621, y = 2.6835, z = 0) show inconsistent RSSI values and differing access point counts. These variations arise from environmental factors, such as multipath propagation and signal attenuation, as well as device-specific attributes, including hardware and antenna design differences [40]. In this study, we removed duplicates with fewer detected access points, retaining those with the most comprehensive access point coverage to improve data quality and enhance model accuracy.

TABLE 4. Variability in detected access points and RSSI values at identical coordinates using the Galaxy Tab S7.

# Detected Access Points	Access Point IDs	Rectified RSSI Values (dBm)
1	AP0	24
3	AP0, AP18, AP119	26, 30, 5
6	AP0, AP1, AP2, AP3 AP4, AP5	24, 48, 16, 15, 43, 42

Table 5 presents the distribution of valid fingerprint samples collected from different devices, along with the number of duplicated coordinates. It also shows the allocation

of these samples into the training and testing datasets, highlighting the total counts for each category.

TABLE 5. Distribution of fingerprint samples, duplicated coordinates, and dataset allocation per device.

Devices	# Valid Fingerprints	# Duplicated Coordinates	# Samples	
			Training	Testing
Galaxy S20	1828	69	1339	420
Galaxy S7	1756	0	1336	420
POCO X3	1747	0	1318	429
Galaxy Tab S7	1835	55	1349	431
Galaxy A12	1733	0	1313	420
Total	**8899**	**124**	**6655**	**2120**

Notably, all duplicated coordinates for the Galaxy S20 are in the training dataset, whereas the Galaxy Tab S7 shows 28 duplicates in the training set and 27 in the testing set.

B. CNN–LSTM MODEL

The CNN–LSTM model employs image-based representations of Wi-Fi signal data to localize a device within an indoor environment. Each image, referred to as a fingerprint, encodes the signal strengths provided by various access points at a given location. In this representation, each pixel corresponds to the RSSI value from a particular access point within the deployment area. The model uses normalized RSSI values ranging from 0 (no signal) to 1 (strongest signal), where black pixels indicate absent signals (no access point detected) and lighter shades correspond to stronger signals. Fingerprint images are constructed by arranging the RSSI values (e.g., 310 values per fingerprint) into a fixed 10×31 format, enabling the model to process them as images. Despite unknown access point locations and differences in the spatial coverage of the access points, the CNN identifies underlying patterns by focusing on non-zero pixel values. Since each physical location has a unique combination of signal strengths from nearby access points, these grayscale patterns form distinctive visual fingerprints. As illustrated in Figure 5, a sample fingerprint image demonstrates how the model spatially organizes the RSSI values, providing the CNN–LSTM model with a structured and learnable representation of the wireless signal environment.

The proposed CNN–LSTM architecture explicitly addresses device heterogeneity and noise in RSSI-based indoor localization by integrating spatial and sequential modeling mechanisms. Each RSSI fingerprint, structured as a two-dimensional grid of signal strengths, encodes spatial patterns that uniquely characterize a location. The CNN processes these fingerprint matrices, with its convolutional kernels learning localized co-occurrence patterns and spatial regularities, thereby reducing dimensionality and suppressing noise. Although the dataset lacks explicit temporal sequences, the LSTM captures structured inter–AP dependencies by treating each fingerprint as an ordered sequence of AP signals. The LSTM models non-linear correlations, stable relative signal strengths, and co-located propagation behaviors while filtering transient fluctuations caused by interference, multipath fading, or hardware variability. These inter–AP relationships often reflect shared propagation paths or co-located APs, improving localization accuracy and enabling generalization to unseen devices. Placing the LSTM after the CNN ensures that correlation modeling operates on refined spatial embeddings rather than sparse, noisy raw inputs, reducing complexity and mitigating overfitting. In this configuration, the CNN refines local spatial features, and the LSTM contextualizes and smooths them to produce robust, device-invariant representations suitable for cross-device localization.

Although RSSI signals are not inherently visual and lack temporal dependencies, arranging multi-access-point fingerprints into structured two-dimensional matrices provides a consistent spatial representation suitable for convolutional processing [12], [13]. This transformation enables the CNN to extract localized co-occurrence patterns and device-invariant features, suppressing noise and reducing dimensionality compared to flat vector representations. The LSTM subsequently interprets each fingerprint as an ordered sequence of access point values, capturing stable inter-AP dependencies and co-located propagation behaviors while filtering transient fluctuations caused by environmental factors or device variability.

By placing the LSTM after the CNN, correlation modeling operates on refined spatial embeddings rather than sparse, noisy raw inputs, enhancing robustness and mitigating overfitting. Together, the CNN–LSTM components produce device-invariant representations that capture both local spatial features and global inter-AP relationships, improving cross-device localization accuracy. This configuration allows the model to discern how signal strengths at one location relate to those at other locations within the deployment area, despite the lack of inherently visual information and temporal sequences in the RSSI signals.

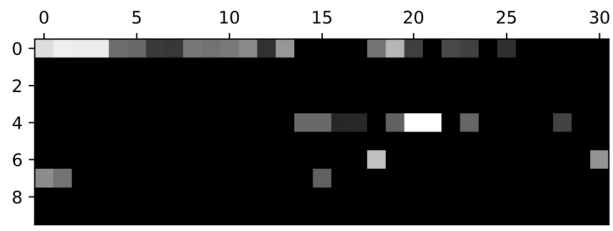

FIGURE 5. Example of a fingerprint image constructed from Wi-Fi RSSI values.

The CNN–LSTM architecture consists of a convolutional layer with 32 filters, followed by an LSTM layer with 32 neurons, with an intermediate reshape layer to match the LSTM input requirements. A flattening layer connects to two fully connected layers with 128 and 64 neurons, each followed by dropout to mitigate overfitting. The final output layer contains three neurons with a linear activation function for coordinate prediction. After training, the output layer is removed, and the 64-neuron dense layer is repurposed as the

new feature output. These features are subsequently passed to a linear regression model and a k-NN model for final location estimation. Figure 6 illustrates the complete architecture.

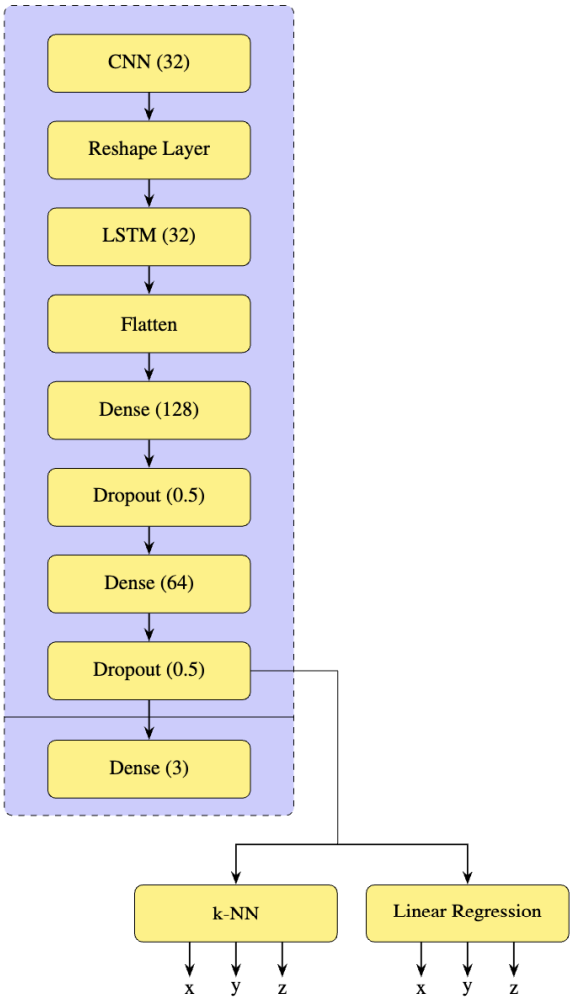

FIGURE 6. The architecture of the CNN-LSTM framework.

This study developed and validated a CNN–LSTM framework to address device heterogeneity in RSSI-based indoor positioning. Future work should investigate advanced deep learning architectures, particularly attention mechanisms and transformer-based models [41], which have the potential to capture richer inter-AP dependencies and long-range contextual relationships. Integrating such approaches into the proposed framework may further improve cross-device generalization under heterogeneous multi-device conditions.

1) MODEL IMPLEMENTATION

The proposed model is implemented in Python using the TensorFlow framework, with Scikit-learn employed for performance evaluation and additional machine learning algorithms. The model is trained for 200 epochs using the Adam optimizer and a mean squared error loss function. The hyperparameters used for model tuning are summarized in Table 6.

TABLE 6. CNN-LSTM model's hyperparameters.

Layer	Filters/ Neurons	Kernel Size	Stride	Activation
CNN	32	2×2	1	ReLU
Reshape (30, -1)	-	-	-	-
LSTM	32	-	-	Tanh
Dense	128	-	-	ReLU
Dropout (0.5)	-	-	-	-
Dense	64	-	-	ReLU
Dropout (0.5)	-	-	-	-
Dense	3	-	-	Linear

2) PERFORMANCE METRIC

The effectiveness of the proposed indoor positioning system is evaluated using the three-dimensional (3D) mean positioning error as the primary performance metric. This metric quantifies the Euclidean distance between predicted and actual locations in three-dimensional space, providing a measure of the model's accuracy. The formula for the mean 3D positioning error is given by: (ϵ_{3D}) is expressed in Equation 4:

$$\epsilon_{3D} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2} \tag{4}$$

The ϵ_{3D} term quantifies the mean 3D positioning error, where (x_1, y_1, z_1) denotes the predicted coordinates and (x_2, y_2, z_2) represents the actual coordinates in the three-dimensional space.

IV. EXPERIMENTS AND RESULTS

This section presents a comprehensive evaluation of the proposed CNN–LSTM-based indoor positioning framework using the TUJI1 dataset. The experiments aim to (i) assess the contributions of individual architectural components, input representations, and hyperparameter settings to the overall performance of the framework, and (ii) analyze the impact of device heterogeneity on IPS effectiveness.

To achieve this, two types of experiments are designed:

1) **Ablation Studies:** These experiments provide insights into the design and configuration of the proposed framework.
   - **Model Architecture Ablation:** Compares MLP, CNN, LSTM, and CNN–LSTM architectures to quantify the benefits of combining spatial and sequential feature learning.
   - **Hyperparameter Ablation:** Investigates how variations in batch size, number of epochs, and validation split ratio affect convergence and accuracy across different CNN–LSTM configurations (standalone, with linear regression, with k-NN regression).
2) **Primary Experiments:** These experiments investigate the impact of device heterogeneity on positioning performance under three scenarios:
   - **Device-Specific Evaluation:** Each device is used independently for training and testing,

allowing assessment of device-specific localization performance.

- **Combined Device Training and Testing:** The model is trained and evaluated on data from all devices to measure overall robustness in heterogeneous environments.
- **Cross-Device Generalization:** One device is excluded during training and later used solely for testing,, evaluating the model's ability to generalize to unseen devices.

Collectively, these experiments provide a comprehensive evaluation of the proposed framework, demonstrating its accuracy and robustness under real-world device heterogeneity, and highlighting the effects of architectural design, training configurations, and regression strategies on IPS performance.

A. ABLATION STUDIES

This section presents two ablation studies designed to provide a deeper understanding of the proposed framework. First, a model architecture ablation assesses the contributions of CNN, LSTM, and their hybrid design in mitigating device heterogeneity. Additionally, an MLP is examined to evaluate whether convolutional kernels capture inter-AP dependencies more effectively than MLPs, even with artificially structured spatial inputs. Second, a hyperparameter ablation investigates the effects of batch size, number of epochs, and validation split ratio on model convergence and positioning accuracy.

1) MODEL ARCHITECTURE ABLATION

To evaluate the contribution of each component in the proposed framework, a model architecture ablation study was conducted. Four architectures were assessed: a fully connected MLP, a CNN, an LSTM, and the proposed hybrid CNN–LSTM. This study aimed to isolate the impact of spatial feature extraction (via CNN) and sequential correlation modeling (via LSTM) on mitigating device heterogeneity in indoor positioning systems.

All models were trained and tested using identical hyperparameters to ensure a fair comparison: 100 epochs, a 0.1 validation split, and a learning rate of 0.001. The architectures for MLP, CNN, and LSTM are detailed in Table 7, while the CNN–LSTM configuration is shown in Table 6. For the MLP baseline, raw 1D RSSI vectors were used as input, whereas CNN, LSTM, and CNN–LSTM models operated on preprocessed 2D fingerprint representations of size 10×31. Notably, no additional machine learning models, such as linear regression or k-nearest neighbors, were employed; location predictions were generated directly by the deep learning models.

Table 8 summarizes the 3D positioning errors obtained for each architecture. The MLP model struggled to learn robust mappings from high-dimensional, noisy RSSI data, resulting in a mean error of 8.35 m. CNN and LSTM individually improved accuracy (3.86 m and 3.91 m, respectively) by capturing local co-occurrence patterns among access points or modeling dependencies in RSSI sequences. The CNN–LSTM hybrid achieved the lowest error of 2.70 m, demonstrating that combining spatial feature extraction with sequential dependency learning produces richer, device-invariant representations and enhances cross-device localization performance.

TABLE 7. Architecture configuration of the MLP, CNN, and LSTM models used in the model architecture ablation experiment.

Model	Layer	Filters/Neurons	Activation
CNN	CNN	32	ReLU
	CNN	16	ReLU
	Flatten	-	-
	Dense	128	ReLU
	Dropout (0.2)	-	-
	Dense	64	ReLU
	Dropout (0.2)	-	-
	Dense	3	Linear
LSTM	LSTM	32	Tanh
	LSTM	16	Tanh
	Flatten	-	-
	Dense	128	ReLU
	Dropout (0.2)	-	-
	Dense	64	ReLU
	Dropout (0.2)	-	-
	Dense	3	Linear
MLP	Dense	128	ReLU
	Dropout (0.2)	-	-
	Dense	64	ReLU
	Dropout (0.2)	-	-
	Dense	3	Linear

These results confirm that CNN and LSTM play complementary roles: the CNN identifies spatial patterns in RSSI fingerprints, while the LSTM refines these features by capturing inter–AP relationships and filtering transient noise. The synergy between these modules explains the superior performance of the CNN–LSTM architecture under heterogeneous device conditions.

Importantly, although RSSI measurements do not inherently contain visual information—particularly because the exact locations of access points are unknown and cannot be directly mapped to the deployment area—representing multi–access-point readings as grayscale images improves performance. This structured spatial arrangement allows CNN layers to consistently learn device-invariant patterns. Such spatially localized dependencies are difficult to capture using flat, fully connected networks.

TABLE 8. The 3D positioning error achieved by different model architectures in the model architecture ablation study, demonstrating the effectiveness of combining CNN and LSTM modules compared to standalone models and the MLP baseline.

Model	ϵ_{3D} $[m]$
MLP	8.35
CNN	3.86
LSTM	3.91
CNN-LSTM	**2.70**

2) HYPERPARAMETERS ABLATION

Following the architecture ablation study, a second set of experiments was conducted to evaluate the influence of

key training hyperparameters on the overall performance of the proposed framework. Specifically, the effects of batch size, number of epochs, and validation split ratio were analyzed in terms of model convergence and positioning accuracy. Table 9 presents the values employed for these hyperparameters. These hyperparameters directly influence the stability of weight updates, the network's generalization ability, and, ultimately, the quality of the learned feature representations.

To ensure a comprehensive evaluation, three variants of the proposed feature extraction backbone were considered, as illustrated in Figure 6:

(i) CNN–LSTM (Standalone): The extracted features are directly mapped to the target coordinates through the final dense layer.

(ii) CNN–LSTM + Linear Regression: A linear regression model is employed as the final estimator to refine the feature-to-coordinate mapping.

(iii) CNN–LSTM + k-NN Regression: The learned feature embeddings are used as input to a k-NN regressor for final position estimation.

TABLE 9. Hyperparameter values explored in the hyperparameter ablation study.

Hyperparameter	Values
Epochs	10, 30, 50, 100
Validation split size	0.1, 0.2, 0.3
Batch size	1, 16, 32, 64

Additionally, the CNN–LSTM architecture employed in this experiment is identical to the configuration presented in Table 6. The k-NN model utilized the Manhattan distance metric, with k set to 10.

The results indicate that the most effective hyperparameter configuration—comprising a batch size of 16, 30 training epochs, and a validation split of 0.1—achieved the lowest mean 3D positioning error, as summarized in Table 10. Figure 7 further illustrates model performance under this configuration, showing how the mean positioning error (3D) varies across different batch sizes and epoch counts for all three model variants.

Notably, the standalone CNN–LSTM model performs poorly when the batch size is set to 1, in contrast to its regression-based counterparts, which exhibit more stable accuracy across configurations. This trend persists when the validation split is increased to 0.2 (Figure 8), with all models maintaining consistent relative performance. However, when the split ratio reaches 0.3 (Figure 9), the CNN–LSTM with linear regression shows a marked performance decline at a batch size of 1 and 30 epochs. Interestingly, its performance substantially improves at 100 epochs under the same batch size, suggesting that extended training mitigates the instability introduced by small batch sizes.

The standalone CNN–LSTM model is trained end-to-end. However, when the batch size is set to m = 1, the stochastic gradient exhibits high variance, causing instability

TABLE 10. Optimal ϵ_{3D} results for different CNN-LSTM variants under the identified best hyperparameter configuration (batch size 16, epoch count 30, validation split 0.1).

Model	ϵ_{3D} $[m]$
CNN-LSTM	2.29
CNN-LSTM with Linear Regression	2.20
CNN-LSTM with k-NN	2.20

during optimization across both the convolutional and recurrent pathways (via backpropagation through the LSTM). This instability produces noisier feature representations and weakens generalization. Prior research has shown that the signal-to-noise ratio of the gradient and the gradient noise scale jointly determine the optimal batch size. Extremely small batches increase update variance, narrow the range of stable learning rates, and bias training toward sharper minima that yield inferior test performance [42], [43].

In contrast, the CNN–LSTM + linear regression and CNN–LSTM + k-NN variants decouple representation learning from the final estimation process. The linear regression head—trained with explicit regularization—and the non-parametric k-NN regressor—which averages over neighbors during inference—act as smoothing and regularizing mechanisms applied to the learned embeddings. This design improves stability and performance, particularly when the encoder is trained with m = 1. Such a fixed-features + simple head configuration is well-documented to provide more stable transfer and evaluation behavior compared to fully end-to-end optimization, especially under challenging or noisy training conditions.

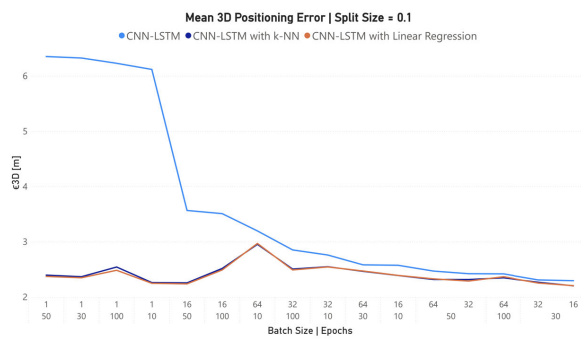

FIGURE 7. Mean 3D positioning error for CNN-LSTM and its variants across batch sizes and epochs (split =0.1).

B. PRIMARY EXPERIMENTS

In this section, three experimental scenarios are designed to evaluate the impact of device heterogeneity on the performance of indoor positioning systems. First, the device-specific evaluation trains and tests the model independently for each device to assess device-level localization accuracy. Second, the combined device training and testing experiment uses data from all devices simultaneously to evaluate overall robustness in a heterogeneous environment. Third, the cross-device generalization experiment excludes

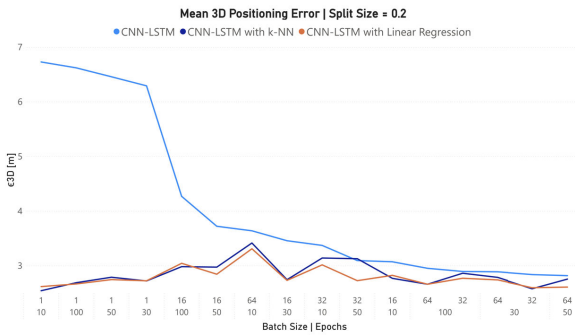

FIGURE 8. Mean 3D positioning error for CNN-LSTM and its variants across batch sizes and epochs (split =0.2).

one device during training and later evaluates the model on that unseen device to measure its generalization capability across different hardware configurations.

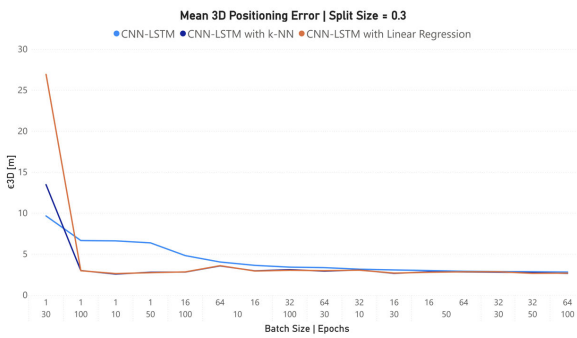

FIGURE 9. Mean 3D positioning error for CNN-LSTM and its variants across batch sizes and epochs (split =0.3).

1) DEVICE-SPECIFIC EVALUATION

In this experiment, each device is trained and evaluated independently to assess its standalone performance, isolated from the influence of other devices. This setup enables a detailed examination of the system's behavior when optimized for a single device's signal characteristics and data distribution. By isolating each device, the experiment identifies device-specific strengths, weaknesses, and potential anomalies that may affect overall localization performance. The k-NN model is employed in this evaluation using the parameters summarized in Table 11, which were meticulously optimized through a grid search procedure.

TABLE 11. Configuration parameters of the k-NN model.

Parameter	Value
Neighbors (k)	10
Metric	Sørensen
Weights	Distance

The findings in Figure 10 reveal notable variation in 3D positioning accuracy across devices when trained and tested independently. The Galaxy S20 achieved the lowest positioning error (2.12 m), whereas the Galaxy A12 exhibited the highest (3.32 m), likely due to hardware limitations. These results highlight the influence of device-specific characteristics on IPS performance. Interestingly, despite the POCO X3 and Galaxy A12 exhibiting nearly identical proportions of invalid RSSI samples (93.24% and 92.51%, respectively), their positioning accuracies differ substantially, with ϵ_{3D} values of 2.22 m and 3.32 m. Conversely, the Galaxy Tab S7, which recorded an even higher rate of invalid RSSI values (96.81%), achieved a lower error (2.26 m) than the Galaxy A12. This discrepancy suggests that, when data are collected from a single device, the number of valid access points exerts a limited effect on positioning performance, whereas device hardware quality and signal stability play a more decisive role.

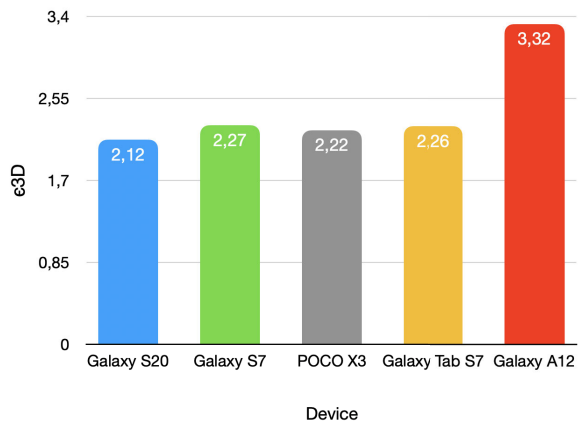

FIGURE 10. The mean 3D positioning error during model formulation and evaluation on a single device.

2) COMBINED DEVICE TRAINING AND TESTING

In this experiment, the proposed CNN–LSTM model is utilized for deep feature extraction, with the resulting representations subsequently used as input to linear regression and k-NN models for location prediction. The model is trained and tested using data from all devices to enable a comprehensive performance assessment under heterogeneous conditions. This setup facilitates a direct comparison with prior studies employing the same dataset, demonstrating the proposed framework's effectiveness in reducing feature dimensionality and alleviating the latency typically associated with exhaustive k-NN searches, particularly when processing large-scale fingerprint datasets.

Table 12 presents the (ϵ_{3D}) of various methods, including our proposed CNN–LSTM model combined with linear regression and k-NN, benchmarked against state-of-the-art techniques reported by Klus et al. [28], [30]. In [28], scripts from Torres-Sospedra et al. [44] were employed for standardized Wi-Fi fingerprint evaluation, implementing 15 methods to reduce the radio map size. These methods are grouped into clustering techniques (e.g., k-means, affinity propagation) and optimization rules (e.g., MoreiraS06, which filters fingerprints based on the strongest access point within

a 6 dB threshold), alongside a baseline k-NN approach. In [30], k-enhanced k-NN with the Intersection-over-Union (IoU) metric was utilized. Our CNN–LSTM framework, when combined with linear regression or k-NN, achieves a superior positioning error of 2.20 m, outperforming leading k-NN-based methods, which report errors ranging from 2.27 m (k-NN with Sørensen) to 2.45 m (k-enhanced k-NN with Sørensen).

Notably, the k-NN Sørensen model with rigorous preprocessing—specifically duplication dropping—achieves a 3D positioning error of 2.42 m, which is nearly identical to the 2.41 m error of the k-enhanced k-NN with the IoU metric. This finding suggests that careful preprocessing can substantially improve conventional k-NN performance, aligning it with more complex enhanced variants. In contrast, the proposed CNN–LSTM framework demonstrates superior accuracy and robustness by effectively leveraging deep feature extraction to reduce dimensionality and mitigate prediction latency. These advantages highlight the model's practical applicability, particularly for large-scale and heterogeneous indoor positioning datasets where computational efficiency and reliability are critical.

TABLE 12. Comparison of performance metrics between proposed and existing methods.

Method	ϵ_{3D} [m]
k-NN with Sørensen [28]	2.27
k-Enhanced k-NN Sørensen [30]	2.45
k-Enhanced k-NN with IOU [30]	2.41
k-NN Sørensen with Preprocessing	2.42
CNN-LSTM Linear Regression	**2.20**
CNN-LSTM k-NN Sørensen	**2.20**

3) CROSS-DEVICE GENERALIZATION

In this experiment, both the k-NN model and the proposed CNN–LSTM architecture were trained using data from all devices except one, which was deliberately excluded during training. During testing, data from all devices—including the excluded one—were used to evaluate the model's ability to generalize to previously unseen devices. This setup provides a rigorous assessment of the system's robustness and adaptability under device heterogeneity. The performance on the excluded device offers valuable insight into the proposed model's capability to accommodate diverse hardware characteristics and maintain reliable positioning accuracy across varying deployment conditions.

Figure 11 illustrates the ϵ_{3D} when the k-NN model is trained on data from all devices except one, which is excluded during the training process, and subsequently tested on all devices. The results reveal that the Galaxy S7 achieved the lowest error of 2.44 m, demonstrating the highest generalization capability. The Galaxy Tab S7, Galaxy S20, and POCO X3 exhibited similar performance, with positioning errors of 2.47 m, 2.49 m, and 2.49 m, respectively. However, the Galaxy A12 recorded the highest error of 2.51 m, indicating that its characteristics might pose challenges for cross-device generalization.

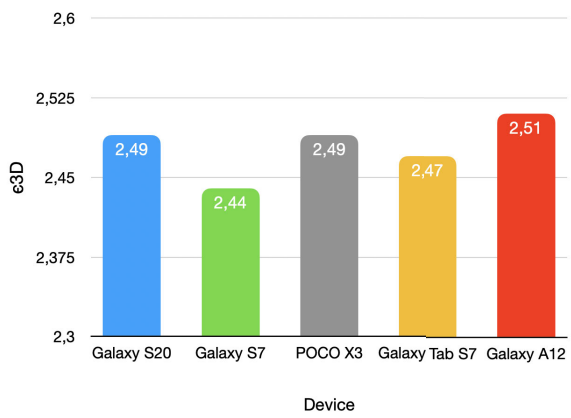

FIGURE 11. The mean 3D positioning error for model development with one device excluded and evaluation across all devices.

The results presented in Table 13 demonstrate the superior performance of the proposed CNN–LSTM model with linear regression compared to the k-NN Sørensen model across all devices. The CNN–LSTM consistently achieved lower ϵ_{3D} values, with reductions ranging from 0.06 m (Galaxy A12) to 0.17 m (Galaxy S20). This improvement highlights the model's robustness and its capacity to mitigate device-specific variability, as evidenced by its narrower error range (2.32–2.43 m) relative to the k-NN Sørensen baseline (2.44–2.51 m). These findings confirm the effectiveness of the CNN–LSTM framework in addressing device heterogeneity and enhancing localization accuracy in RSSI-based indoor positioning systems.

TABLE 13. The mean 3D positioning errors (m) of the proposed method vs. k-NN Sørensen by device.

Devices	CNN-LSTM with Linear Regression	k-NN with Sørensen
Galaxy S20	**2.32**	2.49
Galaxy S7	2.38	**2.44**
POCO X3	2.36	2.49
Galaxy Tab S7	2.41	2.47
Galaxy A12	2.43	2.51

The proposed CNN–LSTM framework demonstrates strong potential for deployment in real-world environments where reliable indoor localization is essential. In smart buildings, it can enhance context-aware automation, energy optimization, and occupant safety, aligning with recent research highlighting the role of machine learning in advancing intelligent building management [45]. In healthcare settings, accurate indoor positioning supports patient tracking, medical asset management, and emergency response. Likewise, in IoT-driven applications—such as robotics, autonomous navigation, and industrial logistics—achieving

device-agnostic localization in dynamic environments is vital for operational efficiency, as emphasized in recent IoT and Wi-Fi-based localization studies [46]. Collectively, these applications illustrate that the proposed CNN–LSTM model extends beyond theoretical evaluation, offering practical benefits for scalable, robust, and cross-device indoor positioning in real-world deployments.

Despite the robustness of the proposed CNN–LSTM framework, several limitations warrant consideration. First, the current evaluation is confined to Wi-Fi RSSI signals. However, integrating complementary technologies such as Bluetooth Low Energy (BLE) and LoRa has been shown to improve positioning robustness across diverse environments [35]. Extending the framework to incorporate multimodal data fusion could therefore enhance resilience under varying propagation conditions. Second, although the model demonstrates real-time feasibility on edge hardware, latency remains a critical concern for large-scale deployments—particularly in scenarios demanding sub-second localization for navigation or safety applications. Future research will explore lightweight model compression, pruning, and adaptive inference strategies to optimize the trade-off between latency and accuracy, thereby improving suitability for time-critical and resource-constrained environments.

4) STATISTICAL SIGNIFICANCE ANALYSIS

A Wilcoxon Signed-Rank Test was conducted to evaluate whether the difference in positioning errors between the CNN–LSTM and k-NN Sørensen methods was statistically significant. This non-parametric paired test was chosen due to the non-normal distribution of the error data and the paired nature of the observations across devices [47]. The null hypothesis (H_0) states that there is no significant difference between the two methods, while the alternative hypothesis (H_a) posits a significant difference in their performance.

The analysis produced a test statistic of 0.0 and a p-value of 0.0625. Although this p-value is slightly above the conventional significance threshold ($\alpha = 0.05$), it suggests a marginal trend toward significance, indicating that the CNN–LSTM model tends to outperform the k-NN Sørensen method. Practically, this finding aligns with the observed results across all five devices, where the CNN–LSTM achieved consistently lower mean 3D positioning errors (2.32–2.43 m) compared to k-NN Sørensen (2.44–2.51 m), as illustrated in Figure 12.

This consistent practical superiority underscores the robustness of the CNN–LSTM framework and its potential for broader applicability across heterogeneous device models [48]. Even marginal accuracy gains (0.1–0.2 m) can substantially enhance performance in precision-critical applications such as medical device localization [49], autonomous navigation [50], and robotic positioning systems [51], where improved localization accuracy directly contributes to operational safety and efficiency. The borderline statistical significance observed in this study likely reflects the limited sample size rather than the absence of a genuine performance advantage.

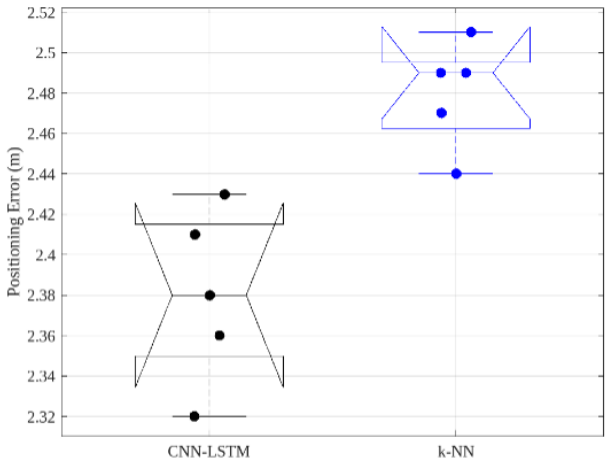

FIGURE 12. Comparison of 3D positioning errors across models.

V. CONCLUSION

This study advances Indoor Positioning Systems by introducing a hybrid CNN–LSTM framework designed to mitigate device heterogeneity through deep feature learning on RSSI fingerprints. By representing RSSI data as structured images, the model effectively captures spatial co-occurrence patterns and inter–access point dependencies, achieving superior accuracy and robustness compared to traditional methods such as k-NN. Cross-device evaluations further demonstrate the framework's ability to generalize across diverse hardware, substantially reducing the impact of device-specific variations. The integration of a linear regression module for efficient coordinate prediction also addresses latency issues inherent to exhaustive comparison-based approaches. Furthermore, by leveraging the TUJI1 dataset, this work contributes to reproducible IPS research and responds to the gap in standardized public benchmarks highlighted by Anagnostopoulos et al. [52].

Despite these advancements, several limitations remain. The high proportion of invalid RSSI samples in certain devices underscores hardware constraints that can degrade accuracy. RSSI signals also remain susceptible to environmental effects such as multipath fading, temporal fluctuations, and interference, limiting reliability in dense or dynamic environments. Moreover, the confined deployment area of the TUJI1 dataset raises questions about scalability to larger and more complex indoor spaces. Since the evaluation was restricted to a single dataset, the generalizability of the proposed framework across broader real-world scenarios warrants further investigation.

Practical deployment introduces additional challenges, including computational overhead on resource-limited devices, inference latency, and the need for recalibration following changes in access-point infrastructure. Privacy concerns inherent to Wi-Fi fingerprinting are further amplified in heterogeneous networks, particularly with the advent

of 6G technologies that enhance localization capabilities while introducing new privacy risks [53]. Addressing these challenges will require secure, efficient, and ethically aligned implementation strategies.

Future research should prioritize multimodal data integration—combining Wi-Fi with complementary sensing modalities such as BLE, inertial sensors, and LoRa—to improve robustness. Exploring transformer-based architectures for richer inter–AP dependency modeling, lightweight model compression for mobile deployment, and privacy-preserving techniques such as federated learning will further advance this line of research. Additionally, validation across broader benchmarks, including UJIIndoorLoc, SODIndoorLoc, and IPIN competition datasets, will strengthen generalizability. Ultimately, this work represents a significant step toward reliable, adaptive, and scalable IPS, paving the way for next-generation localization in smart environments and future 6G ecosystems.

REFERENCES

[1] Y. Gu, A. Lo, and I. Niemegeers, "A survey of indoor positioning systems for wireless personal networks," *IEEE Commun. Surveys Tuts.*, vol. 11, no. 1, pp. 13–32, 1st Quart., 2009, doi: 10.1109/SURV.2009.090103.

[2] F. Zafari, A. Gkelias, and K. K. Leung, "A survey of indoor localization systems and technologies," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 3, pp. 2568–2599, 3rd Quart., 2019, doi: 10.1109/COMST.2019.2911558.

[3] R. Mautz, "Indoor positioning technologies," Ph.D. dissertation, Dept. of Civil, ETH Zurich, Environ. and Geomatic Eng., Zurich, Switzerland, 2012.

[4] R. F. Brena, J. P. García-Vázquez, C. E. Galván-Tejada, D. Muñoz-Rodriguez, C. Vargas-Rosales, and J. Fangmeyer, "Evolution of indoor positioning technologies: A survey," *J. Sensors*, vol. 2017, pp. 1–21, Mar. 2017, doi: 10.1155/2017/2630413.

[5] S. He and S.-H. G. Chan, "Wi-Fi fingerprint-based indoor positioning: Recent advances and comparisons," *IEEE Commun. Surveys Tuts.*, vol. 18, no. 1, pp. 466–490, 1st Quart., 2016, doi: 10.1109/COMST.2015.2464084.

[6] X. Guo, N. Ansari, F. Hu, Y. Shao, N. R. Elikplim, and L. Li, "A survey on fusion-based indoor positioning," *IEEE Commun. Surveys Tuts.*, vol. 22, no. 1, pp. 566–594, 1st Quart., 2020, doi: 10.1109/COMST.2019.2951036.

[7] L. Hao, B. Huang, B. Jia, and G. Mao, "DHCLoc: A device-heterogeneity-tolerant and channel-adaptive passive WiFi localization method based on DNN," *IEEE Internet Things J.*, vol. 9, no. 7, pp. 4863–4874, Apr. 2022, doi: 10.1109/JIOT.2021.3106902.

[8] P. Davidson and R. Piché, "A survey of selected indoor positioning methods for smartphones," *IEEE Commun. Surveys Tuts.*, vol. 19, no. 2, pp. 1347–1370, 2nd Quart., 2017, doi: 10.1109/COMST.2016.2637663.

[9] A. Alarifi, A. Al-Salman, M. Alsaleh, A. Alnafessah, S. Al-Hadhrami, M. Al-Ammar, and H. Al-Khalifa, "Ultra wideband indoor positioning technologies: Analysis and recent advances," *Sensors*, vol. 16, no. 5, p. 707, May 2016, doi: 10.3390/s16050707.

[10] A. A. Abdullah, N. S. Mohammed, M. Khanzadi, S. M. Asaad, Z. K. Abdul, and H. S. Maghdid, "In-depth analysis on machine learning approaches: Techniques, applications, and trends," *ARO-THE SCIENTIFIC J. KOYA Univ.*, vol. 13, no. 1, pp. 190–202, May 2025, doi: 10.14500/aro.12038.

[11] O. Kerdjidj, Y. Himeur, S. S. Sohail, A. Amira, F. Fadli, S. Atalla, W. Mansoor, A. Copiaco, A. Gawanmeh, S. Miniaoui, and D. W. Dawoud, "Uncovering the potential of indoor localization: Role of deep and transfer learning," *IEEE Access*, vol. 12, pp. 73980–74010, 2024, doi: 10.1109/ACCESS.2024.3402997.

[12] Z. Turgut, S. Üstebay, Z. Aydın, and A. Sertbaş, "Deep learning in indoor localization using WiFi," in *Proc. Int. Telecommun. Conf. (ITelCon)*, Istanbul, Turkey, 2018, pp. 101–110, doi: 10.1007/978-981-13-0408-8_9.

[13] Y. Lin, K. Yu, F. Zhu, J. Bu, and X. Dua, "The state of the art of deep learning-based Wi-Fi indoor positioning: A review," *IEEE Sensors J.*, vol. 24, no. 17, pp. 27076–27098, Sep. 2024, doi: 10.1109/JSEN.2024.3432154.

[14] Y. Wei and R. Zheng, "Handling device heterogeneity in Wi-Fi based indoor positioning systems," in *Proc. IEEE Conf. Comput. Commun. Workshops (INFOCOM WKSHPS)*, Toronto, ON, Canada, Jul. 2020, pp. 556–561, doi: 10.1109/INFOCOMWKSHPS50562.2020.9162727.

[15] S. M. Asaad and H. S. Maghdid, "Novel integrated matching algorithm using a deep learning algorithm for Wi-Fi fingerprint-positioning technique in the indoors-IoT era," *PeerJ Comput. Sci.*, vol. 9, p. e1406, May 2023, doi: 10.7717/peerj-cs.1406.

[16] T. Xiang, Y. Sun, and G. Shen, "MoDeFA: Multiobserver and denoising-enhanced fingerprint augmentation for semi-supervised Wi-Fi RSS-based indoor positioning," *IEEE Internet Things J.*, vol. 12, no. 15, pp. 31754–31767, Aug. 2025, doi: 10.1109/JIOT.2025.3573967.

[17] A. Nessa, B. Adhikari, F. Hussain, and X. N. Fernando, "A survey of machine learning for indoor positioning," *IEEE Access*, vol. 8, pp. 214945–214965, 2020, doi: 10.1109/ACCESS.2020.3039271.

[18] X. Feng, K. An Nguyen, and Z. Luo, "A review of open access WiFi fingerprinting datasets for indoor positioning," *IEEE Access*, vol. 12, pp. 167970–167989, 2024, doi: 10.1109/ACCESS.2024.3496561.

[19] K. Davies, I. G. Jones, and J. Shapiro, "A Bayesian approach to dealing with device heterogeneity in an indoor positioning system," in *Proc. 9th Int. Conf. Indoor Positioning Indoor Navigat. (IPIN)*, Jun. 2018, pp. 1–8, doi: 10.1109/ipin.2018.8533730.

[20] A. Sonny, A. Kumar, and L. R. Cenkeramaddi, "A survey of application of machine learning in wireless indoor positioning systems," 2024, *arXiv:2403.04333*.

[21] J. Zhao, N. Frumkin, J. Konrad, and P. Ishwar, "Privacy-preserving indoor localization via active scene illumination," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 1580–1589, doi: 10.1109/CVPRW.2018.00208.

[22] B. A. Dedeturk, B. Kolukisa, and S. Tonyalı, "Privacy-preserving wireless indoor localization systems," *Kocaeli J. Sci. Eng.*, vol. 6, no. 2, pp. 114–128, Nov. 2023, doi: 10.34088/kojose.1098804.

[23] A. Poulose and D. S. Han, "Hybrid deep learning model based indoor positioning using Wi-Fi RSSI heat maps for autonomous applications," *Electronics*, vol. 10, no. 1, p. 2, Dec. 2020, doi: 10.3390/electronics10010002.

[24] Z. Turgut and A. G. Kakisim, "An explainable hybrid deep learning architecture for WiFi-based indoor localization in Internet of Things environment," *Future Gener. Comput. Syst.*, vol. 151, pp. 196–213, Feb. 2024, doi: 10.1016/j.future.2023.10.003.

[25] H. A. Tahir, W. Alayed, W. U. Hassan, and A. Haider, "A novel hybrid XAI solution for autonomous vehicles: Real-time interpretability through LIME–SHAP integration," *Sensors*, vol. 24, no. 21, p. 6776, Oct. 2024, doi: 10.3390/s24216776.

[26] S. M. Asaad, H. S. Maghdid, and Z. K. Abdul, "Hybrid positioning technique using the existing Wi-Fi and LoRa technologies (Wi-Lo)," *Expert Syst. Appl.*, vol. 276, Jun. 2025, Art. no. 127127, doi: 10.1016/j.eswa.2025.127127.

[27] R. F. Qadir, H. S. Maghdid, and A. T. Sabir, "Indoors smartphone positioning enhancement using Wi-Fi and magnetometer," *Polytech. J.*, vol. 13, no. 1, pp. 1–13, Sep. 2023.

[28] L. Klus, R. Klus, E. S. Lohan, J. Nurmi, C. Granell, M. Valkama, J. Talvitie, S. Casteleyn, and J. Torres-Sospedra, "TUJI1 dataset: Multi-device dataset for indoor localization with high measurement density," *Data Brief*, vol. 54, Jun. 2024, Art. no. 110356, doi: 10.1016/j.dib.2024.110356.

[29] J. Torres-Sospedra, R. Montoliu, A. Martínez-Usó, J. P. Avariento, T. J. Arnau, M. Benedito-Bordonau, and J. Huerta, "UJIIndoorLoc: A new multi-building and multi-floor database for WLAN fingerprint-based indoor localization problems," in *Proc. IEEE Int. Conf. Indoor Positioning Indoor Navig. (IPIN)*, Sep. 2014, pp. 261–270, doi: 10.1109/IPIN.2014.7275492.

[30] L. Klus, R. Klus, J. Torres-Sospedra, E. S. Lohan, I. Silva, C. Pendão, and M. Valkama, "Enabling dynamic indoor localization by employing intersection over union as a metric," in *Proc. IEEE 100th Veh. Technol. Conf. (VTC-Fall)*, Sep. 2024, pp. 1–7, doi: 10.1109/VTC2024-FALL63153.2024.10758013.

[31] L. Klus, E. S. Lohan, and M. Valkama, "Employing signal statistics for universal fingerprinting solution," in *Proc. IEEE 100th Veh. Technol. Conf. (VTC-Fall)*, Oct. 2024, pp. 1–7, doi: 10.1109/VTC2024-FALL63153.2024.10757821.

[32] M. Rafiq and A. Jalal, "Wearable sensors-based human locomotion and indoor localization with smartphone," in *Proc. 19th Int. Conf. Emerg. Technol. (ICET)*, Nov. 2024, pp. 1–6, doi: 10.1109/ICET63392.2024.10935024.

[33] Y. Wei, C. Frincu, and R. Zheng, "Informative path planning for location fingerprint collection," *IEEE Trans. Netw. Sci. Eng.*, vol. 7, no. 3, pp. 1633–1644, Jul. 2020, doi: 10.1109/TNSE.2019.2943816.

[34] C. Li, Q. Xu, Z. Gong, and R. Zheng, "TuRF: Fast data collection for fingerprint-based indoor localization," in *Proc. Int. Conf. Indoor Positioning Indoor Navigat. (IPIN)*, Sep. 2017, pp. 1–8, doi: 10.1109/IPIN.2017.8115897.

[35] M. E. Yıldırım, "RSSI based indoor localization with reduced feature dimension," *Balkan J. Electr. Comput. Eng.*, vol. 10, no. 1, pp. 106–109, 2022. [Online]. Available: https://dergipark.org.tr/en/pub/bajece/issue/68157/956866

[36] A. Goldsmith, *Wireless Communications*. Cambridge, U.K.: Cambridge Univ. Press, 2005, doi: 10.1017/cbo9780511841224.

[37] T. S. Rappaport, *Wireless Communications: Principles and Practice*. Cambridge, U.K.: Cambridge Univ. Press, 2024.

[38] L. Lu, Y. Shin, Y. Su, and G. E. Karniadakis, "Dying relu and initialization: Theory and numerical examples," 2019, *arXiv:1903.06733*.

[39] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 448–456.

[40] A. Booranawong, K. Sengchuai, and N. Jindapetch, "Implementation and test of an RSSI-based indoor target localization system: Human movement effects on the accuracy," *Measurement*, vol. 133, pp. 370–382, Feb. 2019, doi: 10.1016/j.measurement.2018.10.031.

[41] H. Zhou, J. Yang, S. Deng, and W. Zhang, "VTIL: A multi-layer indoor location algorithm for RSSI images based on vision transformer," *Eng. Res. Exp.*, vol. 6, no. 1, Mar. 2024, Art. no. 015069, doi: 10.1088/2631-8695/ad213b.

[42] S. McCandlish, J. Kaplan, D. Amodei, and OpenAI Dota Team, "An empirical model of large-batch training," 2018, *arXiv:1812.06162*.

[43] S. Kornblith, J. Shlens, and Q. V. Le, "Do better ImageNet models transfer better?" in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2656–2666, doi: 10.1109/CVPR.2019.00277.

[44] J. Torres-Sospedra, P. Richter, A. Moreira, G. M. Mendoza-Silva, E. S. Lohan, S. Trilles, M. Matey-Sanz, and J. Huerta, "A comprehensive and reproducible comparison of clustering and optimization rules in Wi-Fi fingerprinting," *IEEE Trans. Mobile Comput.*, vol. 21, no. 3, pp. 769–782, Mar. 2022, doi: 10.1109/TMC.2020.3017176.

[45] F. El Husseini, H. N. Noura, O. Salman, and K. Chahine, "Machine learning in smart buildings: A review of methods, challenges, and future trends," *Appl. Sci.*, vol. 15, no. 14, p. 7682, Jul. 2025, doi: 10.3390/app15147682.

[46] M. M. Al-Abossi and B. R. Al-Kaseem, "Indoor localization using Wi-Fi fingerprinting with the Internet of Things: A review," *Al-Iraqia J. Sci. Eng. Res.*, vol. 2, no. 3, pp. 84–93, Sep. 2023, doi: 10.58564/ijser.2.3.2023.90.

[47] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bull.*, vol. 1, no. 6, pp. 80–83, Dec. 1945, doi: 10.2307/3001968.

[48] B. Fariña, D. Acosta, J. Toledo, and L. Acosta, "Improving odometric model performance based on LSTM networks," *Sensors*, vol. 23, no. 2, p. 961, Jan. 2023, doi: 10.3390/s23020961.

[49] D. Benavides, A. Cisnal, C. Fontúrbel, E. de la Fuente, and J. C. Fraile, "Real-time tool localization for laparoscopic surgery using convolutional neural network," *Sensors*, vol. 24, no. 13, p. 4191, Jun. 2024, doi: 10.3390/s24134191.

[50] R. Giri, A. Kumar Sah, and S. Satyal, "Particle filter enhanced by CNN-LSTM for localization of autonomous robot in indoor environment," *Int. J. Eng. Technol.*, vol. 1, no. 2, pp. 1–12, Jun. 2024, doi: 10.3126/injet.v1i2.66654.

[51] N. Chen, H. Wang, G. Fan, D. Yang, and L. Rao, "An end-to-end robotic visual localization algorithm based on deep learning," *J. Sensors*, vol. 2023, no. 1, Jan. 2023, Art. no. 2396911, doi: 10.1155/2023/2396911.

[52] G. G. Anagnostopoulos, P. Barsocchi, A. Crivello, C. Pendão, I. Silva, and J. Torres-Sospedra, "ORDIP: Principle, practice and guidelines for open research data in indoor positioning," *Internet Things*, vol. 30, Mar. 2025, Art. no. 101485.

[53] F. Alhomayani and M. H. Mahoor, "Deep learning methods for fingerprint-based indoor positioning: A review," *J. Location Based Services*, vol. 14, no. 3, pp. 129–200, Jul. 2020, doi: 10.1080/17489725.2020.1817582.

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